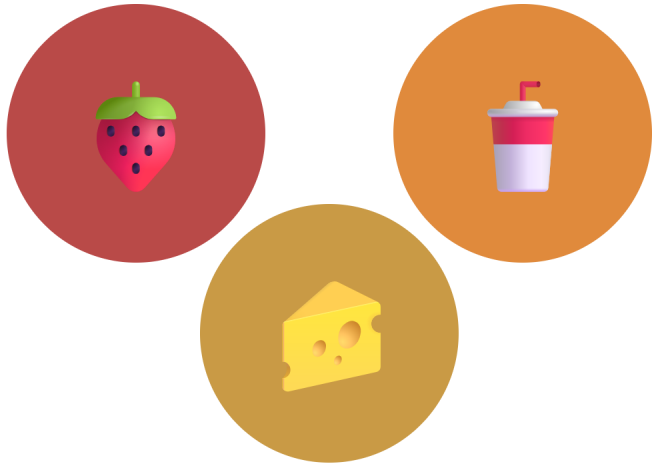
FOOD ADDITIVES

Food Chemistry Seminar

Presented by: Sakshi

B.Sc. Food Science & Technology

Department of Food Science & Technology

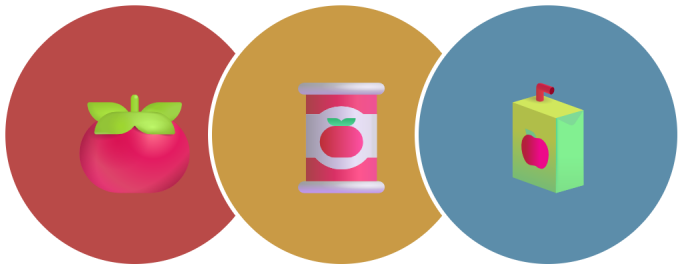


QUALITY • SAFETY • TASTE

Introduction

What are food additives?

- Food additives are substances intentionally added to food in small amounts.
- They help improve taste, colour, texture, quality or shelf life.
- They are used during processing, packaging or storage.

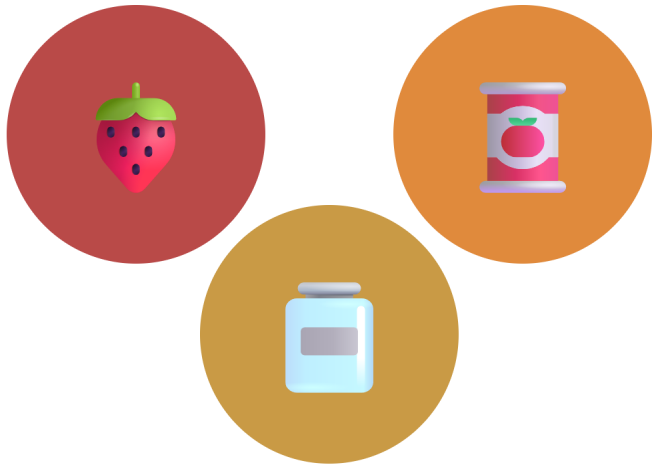


PACKAGED FOODS

Why Are Food Additives Used?

Main purposes

- Preservation — slows food spoilage.
- Colouring — improves or restores appearance.
- Flavouring — improves taste and aroma.
- Sweetening — provides sweetness.
- Emulsification — helps ingredients stay mixed.
- Texture improvement — gives desired consistency.

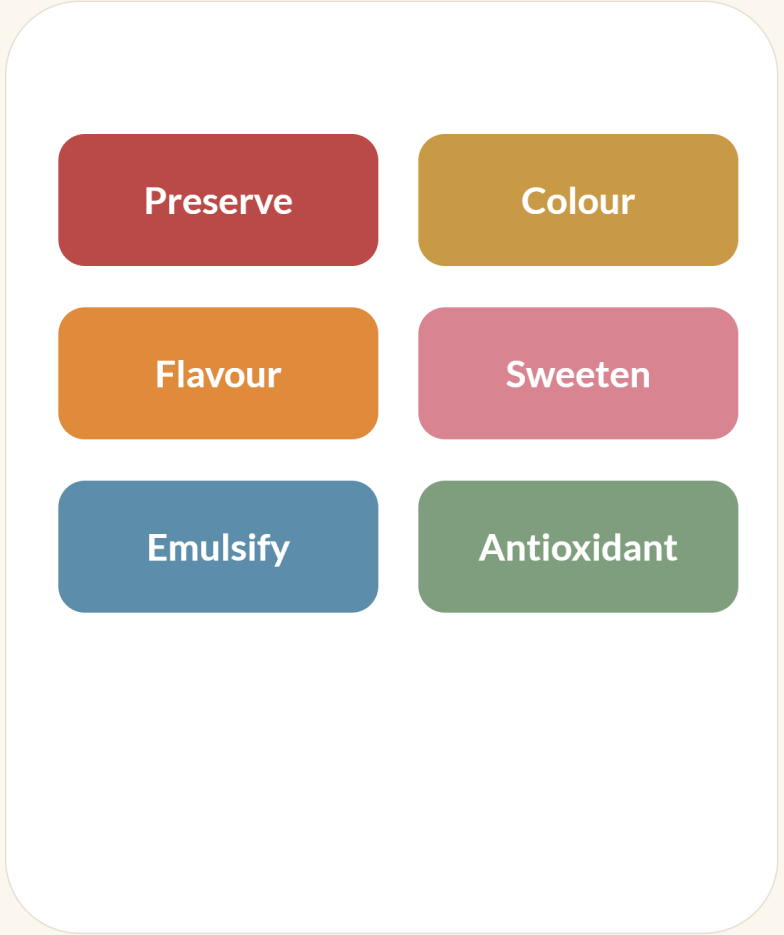


LONGER SHELF LIFE

Classification of Food Additives

Major groups

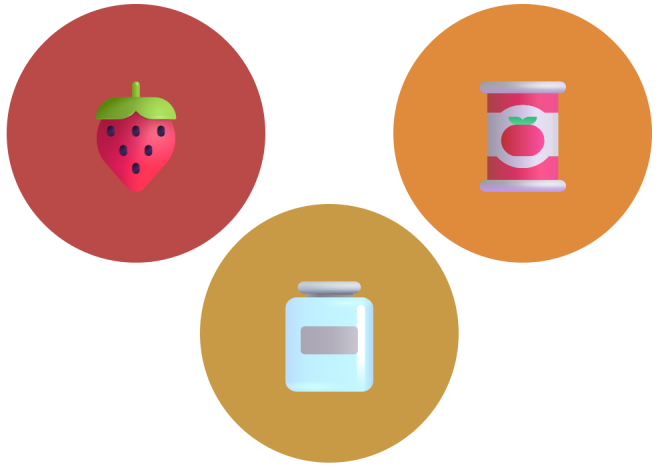
- Preservatives
- Colours
- Flavouring agents
- Sweeteners
- Emulsifiers
- Antioxidants
- Stabilizers & thickeners



Preservatives

Protect food from spoilage

- Preservatives help prevent or slow food spoilage.
- They can increase the shelf life of foods.
- Examples: sodium benzoate, potassium sorbate, salt and sugar.
- Used in beverages, sauces, bakery foods, pickles and jams.

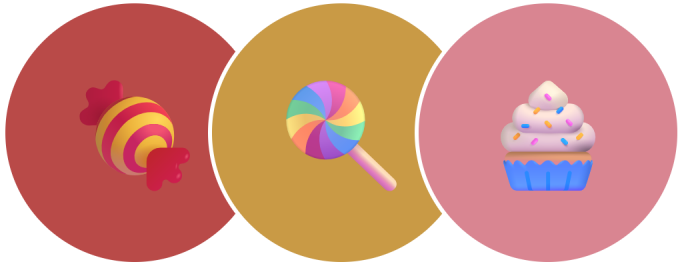


LONGER SHELF LIFE

Food Colours

Improve colour and appearance

- Food colours make food more attractive.
- They may be natural or synthetic.
- Examples: curcumin, caramel colour and beetroot pigments.
- Used in sweets, beverages, ice cream and bakery foods.



COLOUR & APPEARANCE

Flavouring Agents

Improve taste and aroma

- Flavouring agents give or enhance the characteristic flavour of food.
- Examples: vanilla, mint and fruit flavours.
- They are widely used in ice cream, bakery products, beverages and confectionery.

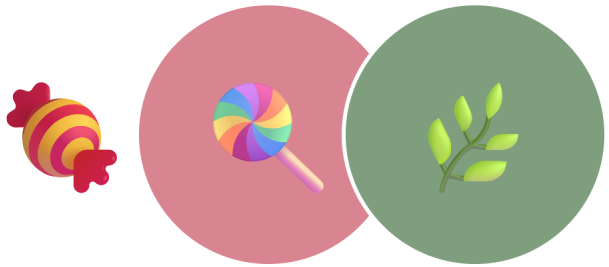


TASTE + AROMA

Sweeteners

Provide sweetness

- Sweeteners are used to make foods or drinks sweet.
- Examples: sugar, glucose, aspartame, sucralose and stevia.
- Commonly used in beverages, candies and bakery products.

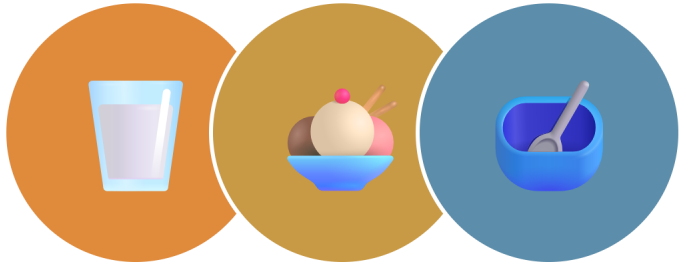


SWEETENERS

Emulsifiers, Stabilizers & Thickeners

Improve texture and consistency

- Emulsifiers help oil and water stay mixed.
- Stabilizers help maintain structure and consistency.
- Thickeners increase the thickness of foods.
- Examples: lecithin, pectin and starch.

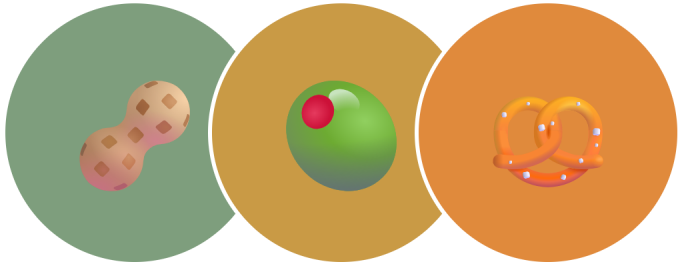


SMOOTH • STABLE • UNIFORM

Antioxidants

Slow oxidation

- Antioxidants help slow oxidation in food.
- Oxidation can cause rancidity and quality loss.
- Examples: ascorbic acid, tocopherols, BHA and BHT.
- Used in oils, snacks and processed foods.

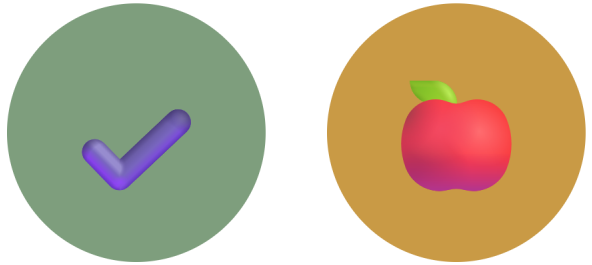


PROTECT FROM OXIDATION

Advantages & Safety

Useful when properly controlled

- Advantages: longer shelf life, improved taste and appearance, better texture and maintained quality.
- Food additives should be used only at permitted levels and according to food regulations.
- Safety depends on the specific additive, amount and intended use.

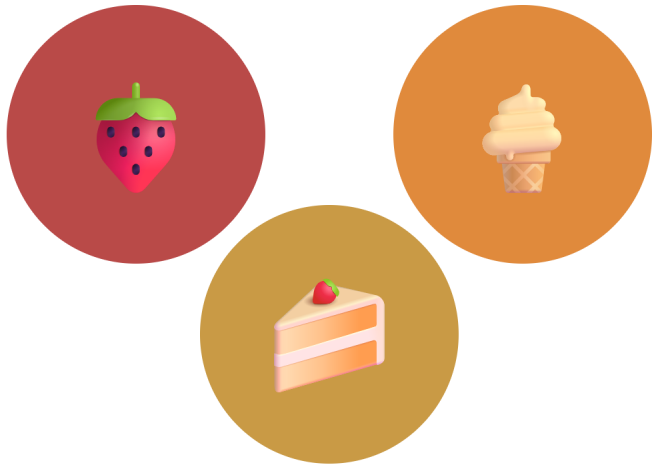


PERMITTED LEVELS

Conclusion

Key message

- Food additives have an important role in modern food processing.
- They can help maintain quality, safety, taste, appearance and shelf life.
- Proper selection and permitted use are important.



QUALITY • SAFETY • TASTE

THANK YOU

Food Additives — Food Chemistry Seminar

Thank You!

Any Questions?

Quality • Safety • Taste • Shelf Life

